Day 003 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Multiply Decimals and Estimate

Standards: 6.NS.B.3

Learning target: I can multiply decimals and use an estimate to place the decimal point.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 9

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01 / Learn

Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate.

Worked example 1

Use the standard algorithm: 36.8 × 7.3. Show your regrouping or division steps.

1. Estimate the size first. Treat the factors as whole numbers: 36.8 becomes 368, and 7.3 becomes 73.

2. Multiply the whole numbers: 368 × 73 = 26864.

3. Count decimal places: 1 in 36.8 plus 1 in 7.3 gives 2.

4. Place the decimal point 2 places from the right in 26864, adding leading zeros if necessary: 268.64. Remove only unnecessary trailing zeros.

Answer: 268.64

Worked example 2

Use the standard algorithm: 4.06 × 0.27. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 481/5000
B. 5481/5000
C. 5481/500
D. 10481/5000

1. Multiply as whole numbers using the standard algorithm.

2. Compute the whole-number product and restore decimal places: 4.06 × 0.27 = 1.0962.

Answer: B - 5481/5000

Day 003 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 19.2 × 1.8. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 46.8 × 2.3. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 22.3 × 3.7. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 7.17 × 7.4. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 9.39 × 5.1. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 3.82 × 3.1. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 3 × 0.62. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 6.92 × 0.8. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 817/125
B. 692/125
C. 567/125
D. 1384/25

09 / Multiple choice

Use the standard algorithm: 7.12 × 0.52. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 2939/625
B. 1689/625
C. 2314/625
D. 4628/125

10 / Constructed response

Ribbon costs $7.07 per meter. What is the exact cost of 0.44 meters? Do not round.

Day 003 | Unit 01 | Answer key

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1. 34.56

Compute the whole-number product and restore decimal places: 19.2 × 1.8 = 34.56.

6.NS.B.3 | Constructed response

2. 107.64

Compute the whole-number product and restore decimal places: 46.8 × 2.3 = 107.64.

6.NS.B.3 | Constructed response

3. 82.51

Compute the whole-number product and restore decimal places: 22.3 × 3.7 = 82.51.

6.NS.B.3 | Constructed response

4. 53.058

Compute the whole-number product and restore decimal places: 7.17 × 7.4 = 53.058.

6.NS.B.3 | Constructed response

5. 47.889

Compute the whole-number product and restore decimal places: 9.39 × 5.1 = 47.889.

6.NS.B.3 | Constructed response

6. 11.842

Compute the whole-number product and restore decimal places: 3.82 × 3.1 = 11.842.

6.NS.B.3 | Constructed response

7. 1.86

Compute the whole-number product and restore decimal places: 3 × 0.62 = 1.86.

6.NS.B.3 | Constructed response

8. B - 692/125

Compute the whole-number product and restore decimal places: 6.92 × 0.8 = 5.536.

6.NS.B.3 | Multiple choice

9. C - 2314/625

Compute the whole-number product and restore decimal places: 7.12 × 0.52 = 3.7024.

6.NS.B.3 | Multiple choice

10. 3.1108 dollars

Compute the whole-number product and restore decimal places: 7.07 × 0.44 = 3.1108.

6.NS.B.3 | Constructed response